



Logic Gates

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Logic Gates

- Basic Digital logic is based on 3 primary functions (the basic gates):

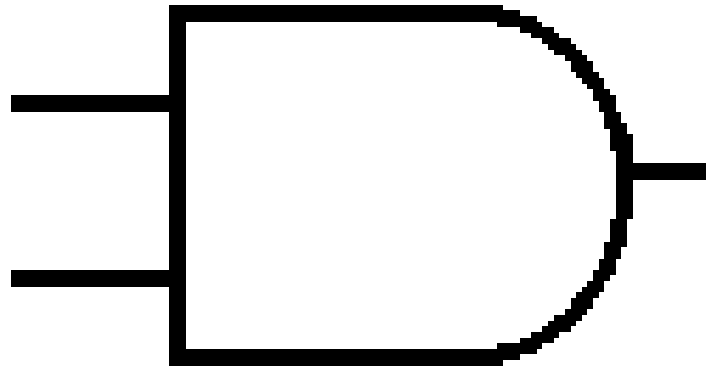
- AND
- OR
- NOT

The AND function

- If all the inputs are high is the output is high
- If any input is low, the output is low
- “If this input AND this input are high, the output is high”

AND Logic Symbol

Inputs

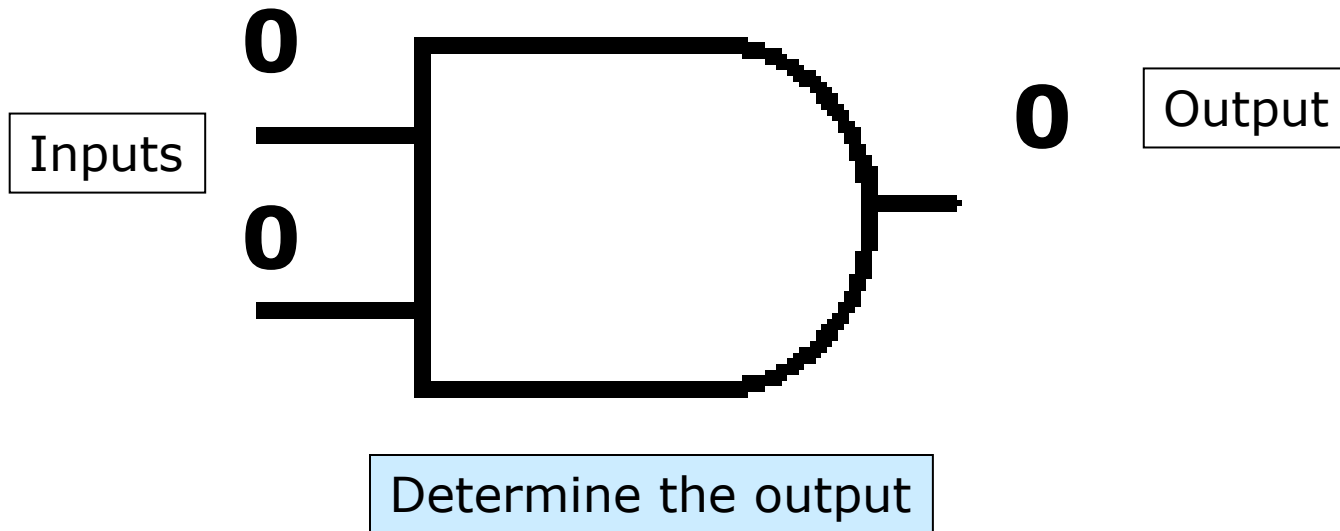


Output

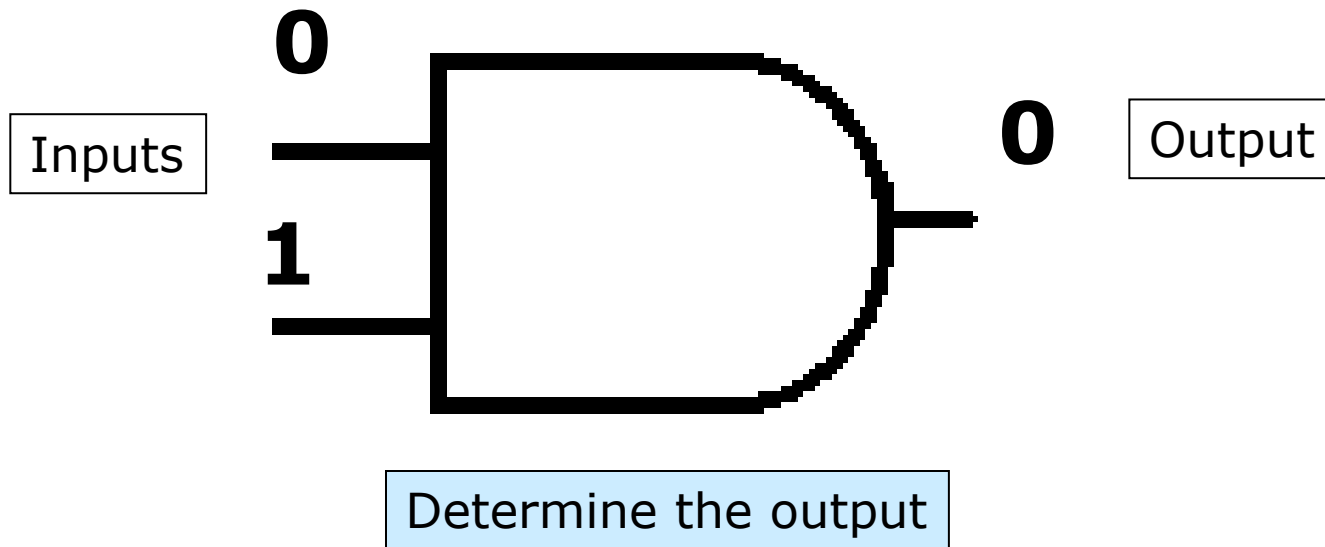
If both inputs are 1, the output is 1

If any input is 0, the output is 0

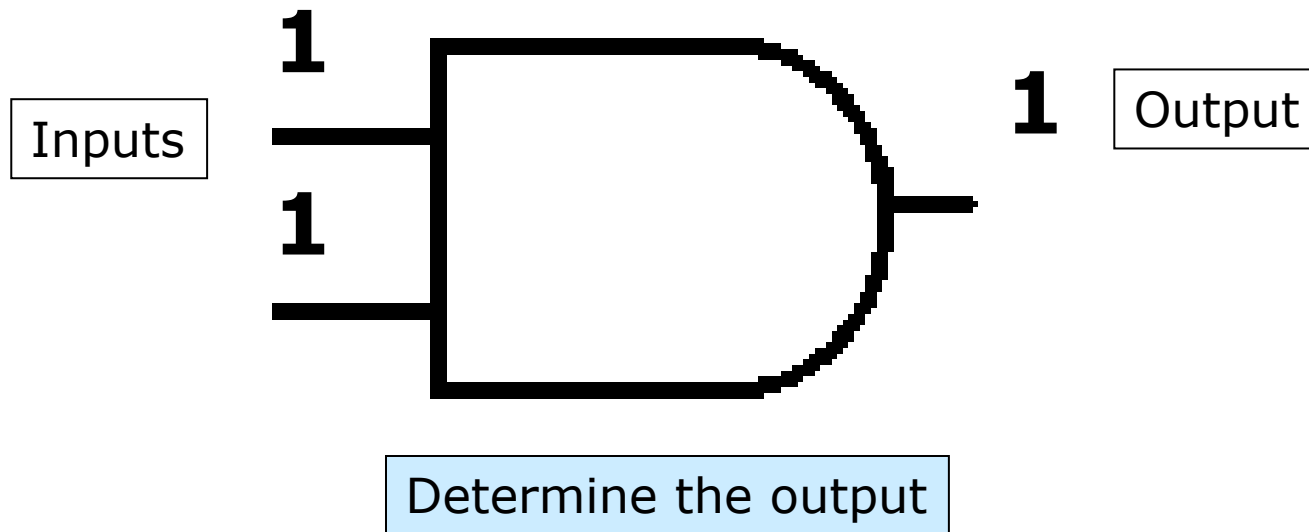
AND Logic Symbol



AND Logic Symbol



AND Logic Symbol



AND Truth Table

- To help understand the function of a digital device, a Truth Table is used:

Every possible
input combination



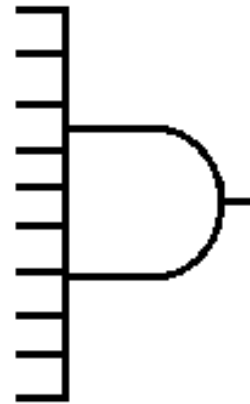
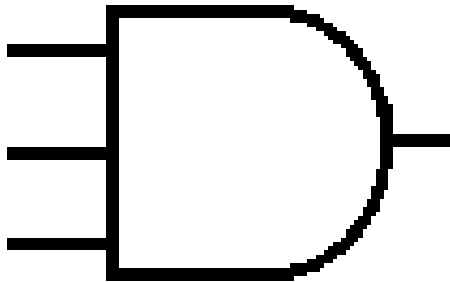
Input		Output
0	0	0
0	1	0
1	0	0
1	1	1

AND Function

AND Gates



It is possible to have AND gates with more than 2 inputs. The same logic rules apply – “if any input...”



The OR function

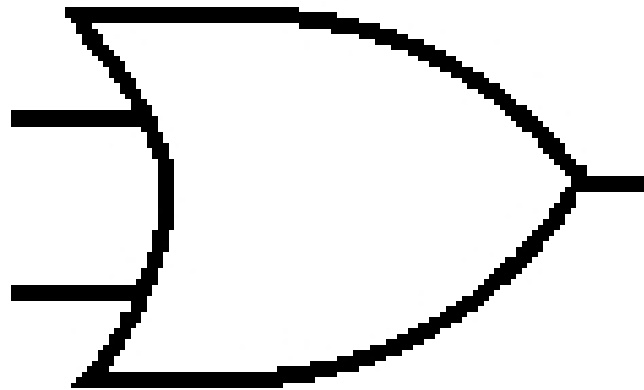
The OR function:

- if any input is high, the output is high
- if all inputs are low, the output is low

- “If this input OR this input is high, the output is high”

OR Logic Symbol

Inputs



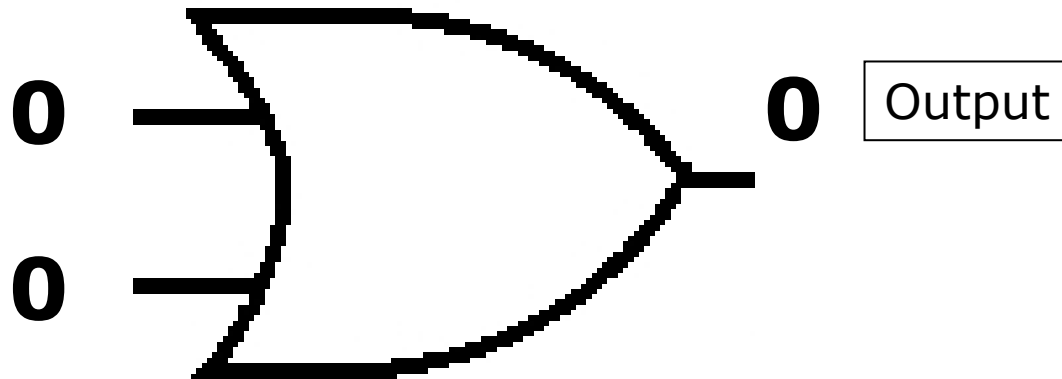
Output

If any input is 1, the output is 1

If all inputs are 0, the output is 0

OR Logic Symbol

Inputs

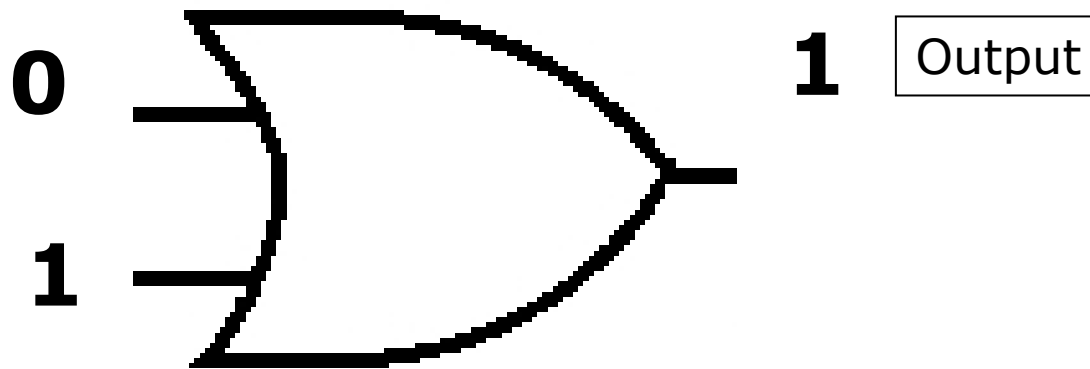


Output

Determine the output

OR Logic Symbol

Inputs



Output

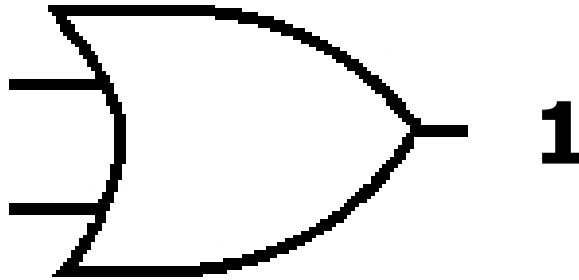
Determine the output

OR Logic Symbol

Inputs

1

1



Output

1

Determine the output

OR Truth Table

- Truth Table

Input		Output
0	0	0
0	1	1
1	0	1
1	1	1

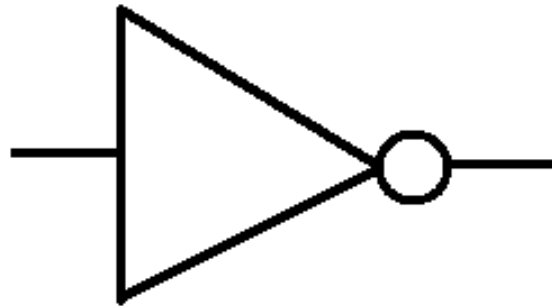
OR Function

The NOT function

- The NOT function:
 - If any input is high, the output is low
 - If any input is low, the output is high
- “The output is the opposite state of the input”
- The NOT function is often called INVERTER

NOT Logic Symbol

Input



Output

If the input is 1, the output is 0

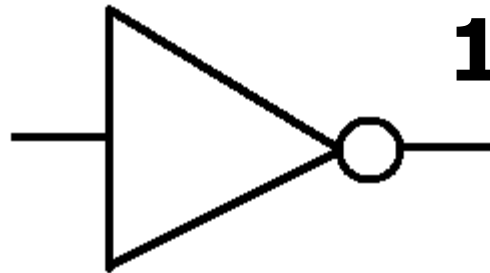
If the input is 0, the output is 1

NOT Logic Symbol

Input

Output

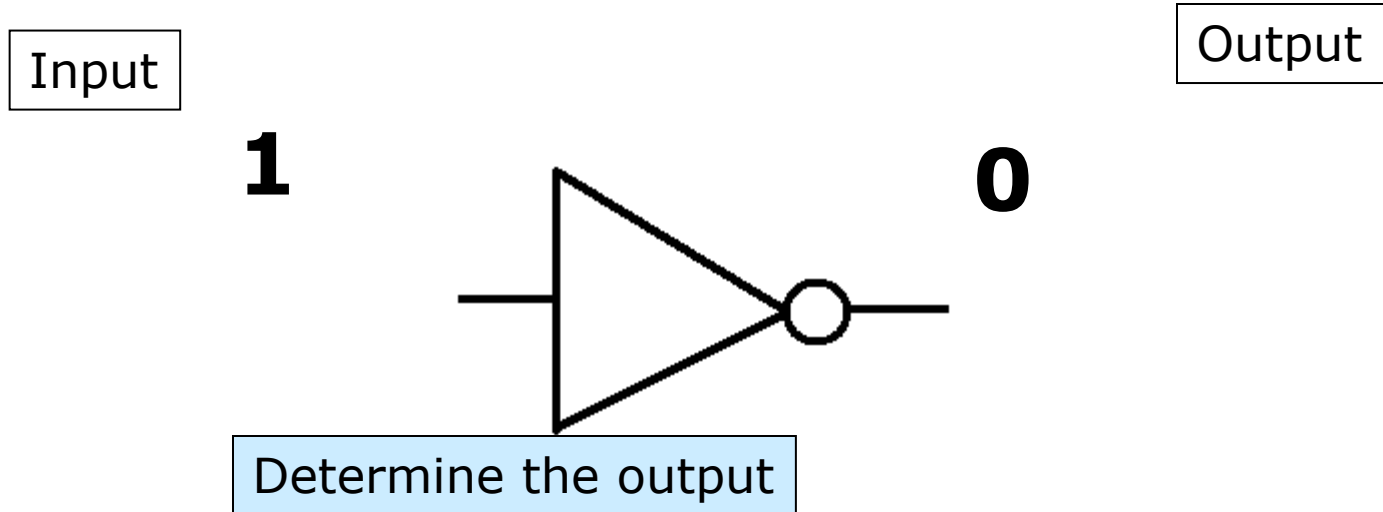
0



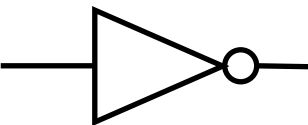
1

Determine the output

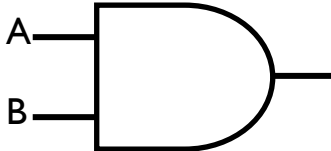
NOT Logic Symbol



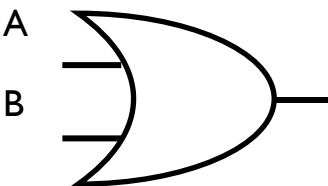
Logic Gates

NOT: In 

A	Out
0	1
1	0

AND: 

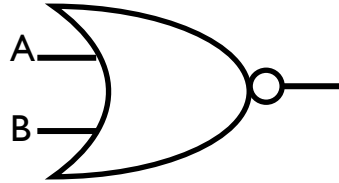
A	B	Out
0	0	0
0	1	0
1	0	0
1	1	1

OR: 

A	B	Out
0	0	0
0	1	1
1	0	1
1	1	1

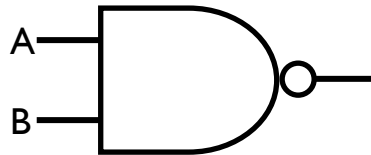
More Logic Gates

NOR:



A	B	Out
0	0	1
0	1	0
1	0	0
1	1	0

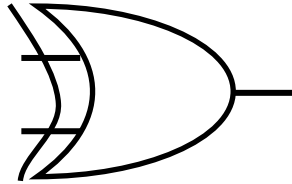
NAND:



A	B	Out
0	0	1
0	1	1
1	0	1
1	1	0

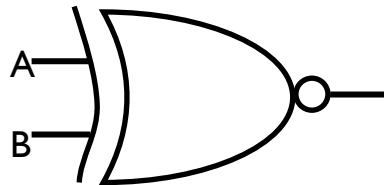
More Logic Gates

XOR:



A	B	Out
0	0	0
0	1	1
1	0	1
1	1	0

XNOR:



A	B	Out
0	0	1
0	1	0
1	0	0
1	1	1

Logic Equations

NOT:

- $\text{out} = \bar{a}$

AND:

- $\text{out} = a \cdot b$

OR:

- $\text{out} = a + b$

XOR:

- $\text{out} = a \oplus b = a\bar{b} + \bar{a}b$

NAND:

- $\text{out} = \overline{a \cdot b}$

NOR:

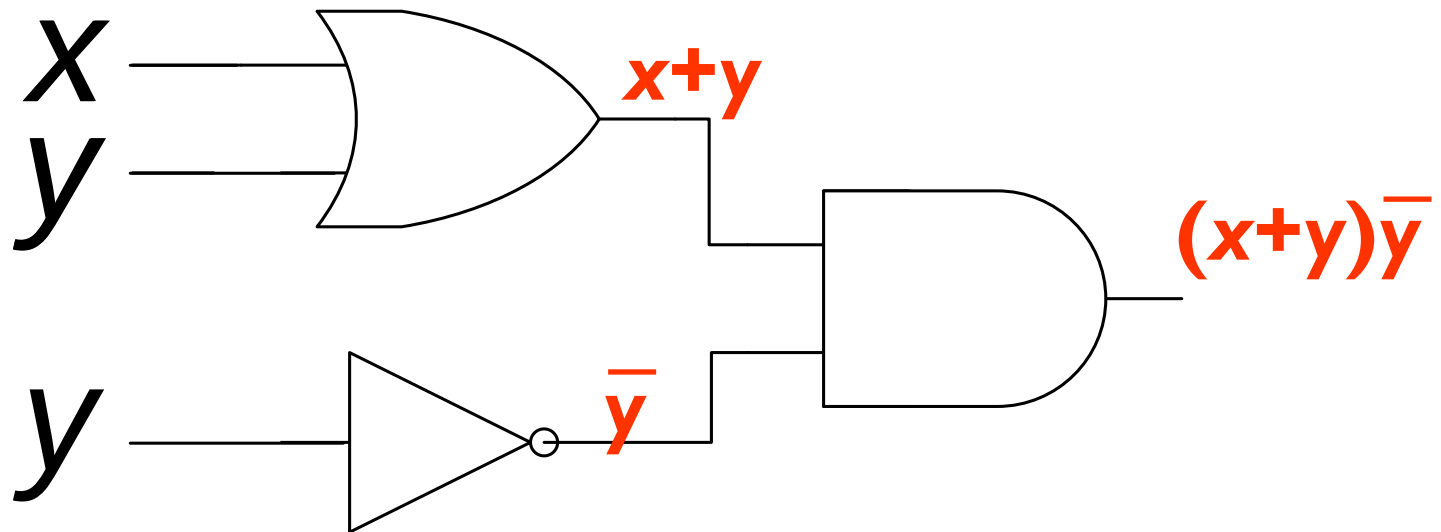
- $\text{out} = \overline{a + b}$

XNOR:

- $\text{out} = \overline{a \oplus b} = ab + \bar{a}\bar{b}$

Converting between circuits and equations

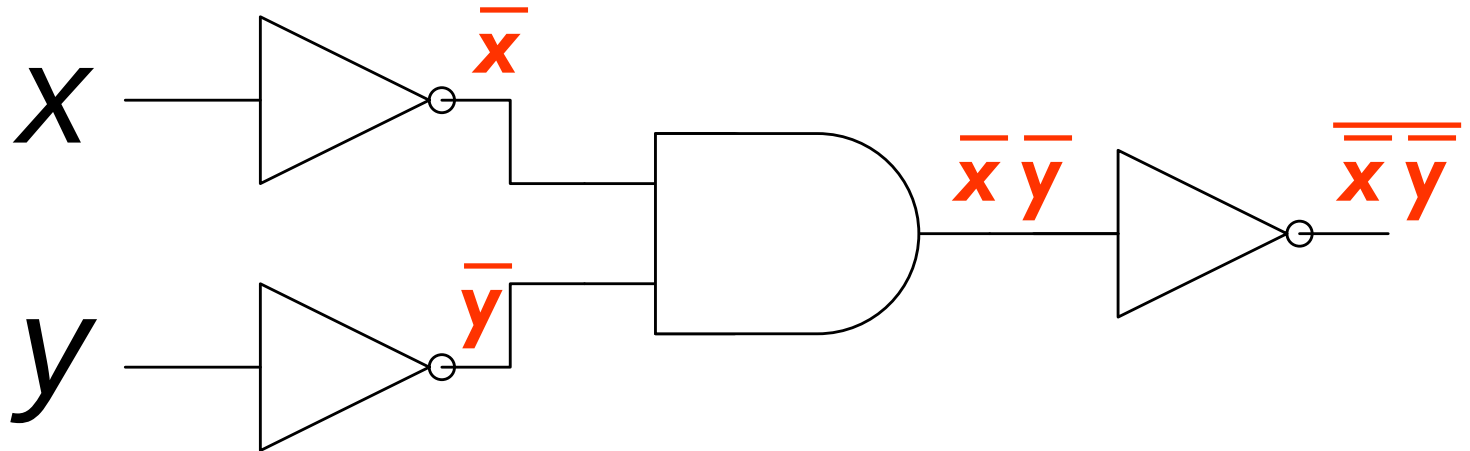
- Find the output of the following circuit



- Answer: $(x+y)\bar{y}$

Converting between circuits and equations

- Find the output of the following circuit

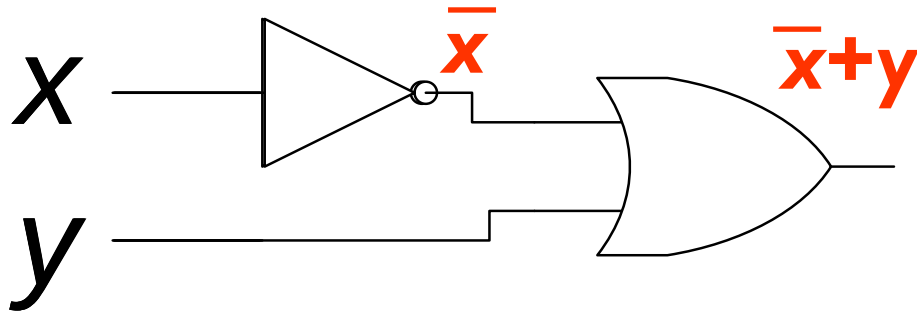


Answer: $\overline{\bar{x}\bar{y}}$

Converting between circuits and equations

- Write the circuits for the following Boolean algebraic expressions

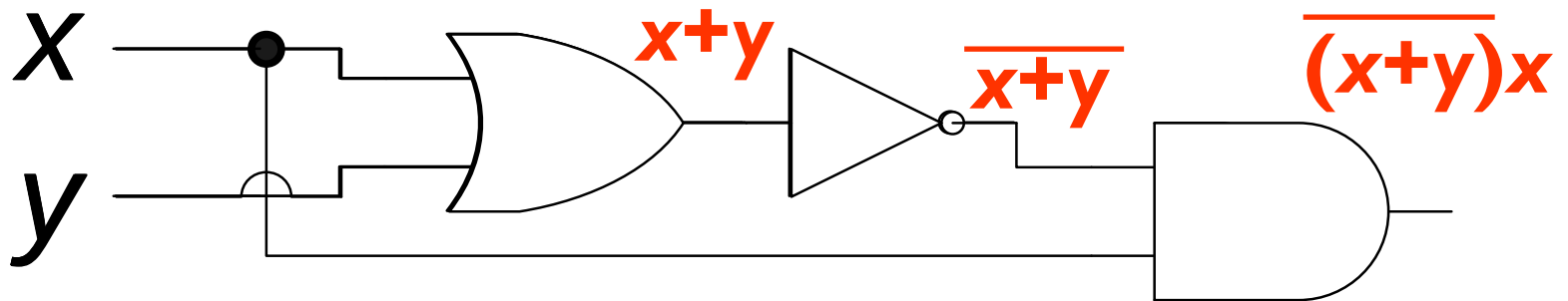
a) $\overline{x} + y$



Converting between circuits and equations

- Write the circuits for the following Boolean algebraic expressions

b) $\overline{(x+y)}x$



Binary Addition

Table 1.5 One-bit adder.

a	b	c_{in}	c_{out}	s
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Binary Addition

Figure 1.2 A 4-bit adder.

